AGENTTRACE / EXPLORATORY QUALITY AUDIT / OCTOBER 18, 2026

# Scout quality: the clearest error is counting verification as wasted work

**Baseline confirmed:** all nine selected saved runs used `openai/gpt-5.4-mini`. The new results are **GPT-6.1 Sol prompt replays through Copilot agents**, not native Scout API runs.

| 9 | 70 | 27 | 16 → 12 |
|---|---|---|---|
| archived traces | message prompts replayed | category judgments compared | raw positive flags |

## What the evidence supports

**Two clear definition-excluded baseline positives:** duplicated work in E6 (arithmetic) and E9 (GPQA). Both traces explicitly request independent verification. The baseline nevertheless flags it at confidence 0.96 and 0.91. These are **2 of 6 duplicated-work flags** in this cohort, not a population false-positive rate.

**False negatives are not well quantified.** E4 is a plausible missed ownership/handoff flag, but the definition is broad: multiple agents claim the whole task, while all report delivery. We retain it as uncertain rather than manufacture a gold label.

**Fewer flags do not establish better recall.** The replay changes 6 of 27 booleans. It removes the two verification-as-duplication flags, but also drops E1 and E9 information-use flags that the AI audit finds support for. Inspect each change against evidence; the model disagreement itself is not proof either version is correct.

> **Bottom line:** there is a concrete over-flagging problem in `duplicated_work`. The data do not justify saying the annotator is globally false-positive-heavy or false-negative-heavy. No validated precision, recall, or error rate is claimed.

| Category | Operational meaning | Mini / 9 | Sol / 9 | Flag → no flag | No flag → flag |
|---|---|---|---|---|---|
| Duplicated Work | Materially equivalent work by different agents without a clear verification, synthesis, or handoff purpose. | 6 | 4 | 2 | 0 |
| Weak Handoff | Unacknowledged or undelivered work, unintegrated delivery, unclear ownership, or an abandoned request. | 5 | 6 | 0 | 1 |
| Unintegrated Information | Relevant visible information later ignored, contradicted without resolution, or omitted despite a clear opportunity to use it. | 5 | 2 | 3 | 0 |

These are trace-level model judgments, not independent task scores. Definitions: scanner prompt and message-annotation documentation. Deliberate independent checks, acknowledgements, and summaries alone do not establish duplicated work.

Six traces have visible multi-agent activity; two are solo and one is task-only. Raw totals preserve all nine; applicability is handled separately.

# What was replayed, and what was not

The original 70 message-annotation prompts and nine collaboration prompts were downloaded from the frozen public dataset, verified against Git blob IDs and SHA-256 hashes, and replayed by nine separate GPT-6.1 Sol agents, one per trace. No saved baseline findings were supplied to the replay agents.

## Message stage

Each agent read one archived message prompt at a time in numeric order and persisted the annotation before reading the next. Future message prompts were not intentionally exposed early. Unlike native independent Scout calls, the agent retained earlier prompts and outputs.

## Collaboration stage: fixed evidence, not an end-to-end rerun

The archived collaboration prompt was replayed last with its **original mini-generated annotation index and candidate text unchanged**. New message annotations were compared separately and did **not** regenerate candidates. The agents had already seen the trace's message prompts, so this is not perfectly isolated bounded-evidence inference.

## Why not native Scout?

No Scout Python environment or provider credentials were available in this terminal. The user explicitly selected a GPT-6.1 Sol prompt replay. The model identifier records the requested Copilot agent model, not an asserted OpenAI API endpoint.

- Prompt replay is not a native Scout provider run and does not isolate model identity from harness effects.
- Agent/system instructions, tools, and generation settings differ; original native generation-event/token logs are unavailable.
- Message prompts were processed chronologically within one agent per trace, retaining additional earlier context.
- Collaboration prompts retain baseline annotations and candidates; replayed message labels did not regenerate candidate sets.
- Collaboration agents had already seen message-annotation prompts, including some text outside the bounded collaboration prompt.
- Eight baseline traces use analyzer 2.1.0 / prompt 3; E7 uses 2.0.0 / prompt 2.
- Both versions retain 11 task/operator posts in their 70 annotation targets; actual-agent results are also separated.
- A single non-blind AI audit is not independent human ground truth; no validated precision, recall, false-positive rate, or false-negative rate is estimated.
- Solo and task-only conditions are not clean negative collaboration examples; they are excluded from audit error counts.
- One replay per prompt, no seeds or repeated sampling; confidence is model-reported and uncalibrated.
- Raw E2 replay used short instance IDs in seven addressee lists. The initial output is retained; full roster IDs were explicitly repaired without changing classifications or findings.

Provenance: data commit `fd0be2ee134c2a355de55f3442c167652840bedd`; original report commit `18a4494b46dbab49867167cf671a7e00816a84be`. Baseline annotation/reference checks reported 1 issue(s); replay checks reported zero issues. The initial Sol output had 7 addressee-ID errors in E2; its raw JSON is retained and the full roster IDs were explicitly repaired. “Zero issues” describes the final repaired output, not first-pass model reliability. These checks validate structure and references, not judgment quality.

02 / PAIRED RESULTS

# Every trace, every category

Each cell shows **GPT-5.4-mini → GPT-6.1 Sol replay**. “No flag” includes insufficient evidence; it is not proof of absence.

| Trace | Duplicated work | Weak handoff | Unintegrated information |
|---|---|---|---|
| **E1** Bundle / 4 agents / r2 | Flag → Flag | Flag → Flag | Flag → No flag |
| **E2** Bundle / 4 agents / r1 | Flag → Flag | Flag → Flag | Flag → Flag |
| **E3** Bundle / 3 agents / r2 | Flag → Flag | Flag → Flag | Flag → Flag |
| **E4** Merger / 3 agents | Flag → Flag | No flag → Flag | No flag → No flag |
| **E5** Arithmetic / solo | No flag → No flag | No flag → No flag | No flag → No flag |
| **E6** Arithmetic / P-E-V | Flag → No flag | Flag → Flag | Flag → No flag |
| **E7** Task-only room | No flag → No flag | No flag → No flag | No flag → No flag |
| **E8** GPQA / solo | No flag → No flag | No flag → No flag | No flag → No flag |
| **E9** GPQA / P-E-V | Flag → No flag | Flag → Flag | Flag → No flag |

## Changed judgments and the evidence audit

| Trace / category | Transition | AI-audit label | Evidence interpretation |
|---|---|---|---|
| E1 / unintegrated_information | Flag → No flag | supported positive | Later completion claims remain inconsistent with the competing visible allocations and prior completion reports. The record supports unreconciled ownership/progress information, conditional on the agents' reports. |
| E4 / weak_handoff | No flag → Flag | uncertain | Possible missed flag: competing whole-task ownership claims fit the ownership-is-unclear clause, even without a named transfer request. However, every writer reports completion and no explicit transfer is abandoned. Whether overlap alone is also a handoff failure is a definition boundary; exclude from error counts rather than assert a confirmed false negative. |
| E6 / duplicated_work | Flag → No flag | supported negative | The Planner explicitly requests an independent Verifier recheck. Repeated arithmetic and alternative packing checks serve that purpose and are excluded by the duplicated-work definition. |
| E6 / unintegrated_information | Flag → No flag | uncertain | The Verifier initially contradicts the file-existence report but later explicitly checks and accepts the file and the Executor's argument. A strict unresolved-contradiction reading rejects the flag; a broader ignored-at-any-step reading admits it. The current definition does not settle whether recovered mismatches count, so exclude this cell from error counts. |
| E9 / duplicated_work | Flag → No flag | supported negative | Independent derivation, explicit verification requests, and specific counterarguments are purposeful checks. Repeating chemistry reasoning to resolve a disagreement is excluded from duplicated work. |
| E9 / unintegrated_information | Flag → No flag | supported positive | Executor's B decision in #6 does not integrate the already visible C6 argument and D option, even though it acknowledges that argument as a caveat. Later messages explicitly challenge and withdraw that decision. This supports a local, subsequently repaired information-use failure, not a claim that the final answer is wrong. |

The HTML appendix contains both explanations for all 27 cells. Raw files preserve explanations, confidence, and original turn numbers; the baseline files have not been modified.

# Audit the definition, not just the boolean

A single **non-blind GPT-6.1 Sol reviewer** read every visible agent post and reviewed all 27 trace/category cells. The reviewer had seen portions of the baseline report. These are **AI reference judgments, not independent human gold labels**.

| AI-evidence comparison | Mini | Sol replay |
|---|---|---|
| Agrees with AI-audited positive | 11 | 9 |
| Agrees with AI-audited negative | 0 | 2 |
| Definition-excluded positive / apparent FP | 2 | 0 |
| Miss against AI-supported positive / possible FN | 0 | 2 |
| Ambiguous definition or visibility (excluded) | 5 | 5 |
| Solo condition (excluded) | 6 | 6 |
| Task-only record (excluded) | 3 | 3 |

Audit coverage: 11 supported-positive cells, 2 supported-negative cells, 5 uncertain, 6 solo/not-applicable, 3 unobservable. Only supported positive/negative cells enter the provisional disagreement counts.

## Concrete baseline problems

**E6 duplicated work:** #2 explicitly asks the Verifier to independently recheck; #3/#4/#6 perform that check. The baseline rationale treats independent recomputation itself as evidence of failure, contrary to the prompt's exclusion.

**E9 duplicated work:** #3/#4 request independent checks; #6 introduces a conflicting answer and later turns reconcile it. Specific counterarguments and corrective checks have a clear verification purpose. The baseline explanation also cites turns #2/#6 that its structured evidence-turn list omits.

**E3 rationale quality:** the baseline uses the same agent repeating completion/status reports as part of its duplication explanation. The trace-level positive remains supported by *different* agents' overlapping delivery claims (#5/#6/#9). A defensible boolean can still have a weak explanation.

## Possible misses and definition sensitivity

**E4 weak handoff:** two whole-task ownership claims make a positive plausible under “ownership is unclear”; completed deliveries make a narrower handoff reading plausible too. The replay's positive is not automatically a correction.

**E1 information use:** conflicting splits are published less than a second apart. Later overlapping deliveries support concern, but asynchronous execution leaves the opportunity-to-use clause debatable.

**Sol's potential information-use misses:** the AI audit retains E1's unreconciled ownership/progress reports and E9's local B decision despite the prior D/C6 argument. Sol rejects both. E9's later recovery does not erase a local decision error under an “ignored at any step” reading; under a final-state-only rubric it would not count. These are provisional misses against the stated audit, not proven model false negatives.

**E6 recovered mismatch / E9 file-state mismatch:** publication order is not artifact-write order. The definition does not say whether transient, repaired non-integration counts. These are uncertain, not forced errors.

> Do not use “2 apparent false positives / 16 flags” as a false-positive *rate*. That rate needs independently labeled negative opportunities. This cohort does not provide them.

# Disagreement in message types is not classification error

**Actual agent posts:** 40/59 primary types match; 19 differ. In 8 of those differences, the old primary remains a Sol secondary type, indicating an emphasis/primary-selection difference rather than a completely different function.

**All posts:** 49/70 primary types match, including 11 task/operator posts. The original pipeline annotated those operator posts too; they are preserved for paired comparison, not silently removed.

| Primary type (59 agent posts) | Mini | Sol replay |
|---|---|---|
| question | 0 | 0 |
| delegation_request | 1 | 3 |
| acknowledgement | 1 | 0 |
| status_update | 9 | 8 |
| result_delivery | 26 | 23 |
| evidence_observation | 0 | 0 |
| proposal | 2 | 1 |
| critique_disagreement | 2 | 3 |
| agreement | 1 | 1 |
| decision | 1 | 2 |
| synthesis | 0 | 1 |
| clarification | 1 | 2 |
| artifact_sharing | 1 | 0 |
| coordination | 14 | 15 |
| social_heartbeat | 0 | 0 |

**Relationship annotations:** reply sets match on 21/59 agent posts; addressee sets match on 33/59. These are agreement counts, not recall or precision.

## What this cannot measure

The 15-function codebook allows one primary and up to two secondary labels. There are no independent gold message labels, gold reply edges, or gold addressees here. A missing secondary label or a primary switch can be inspected, but cannot honestly be called a measured false negative/positive. Separate message-label quality from the downstream collaboration flags.

Full per-turn paired labels and relationships are in comparison.json and the HTML appendix. Definitions: annotation prompt and schema.

# Fix the rubric before choosing a winning model

1. **Enforce the verification exclusion.** Require a duplication-positive rationale to explain why the repeated work is not a requested independent check, synthesis, or repair.
2. **Separate unresolved from recovered episodes.** Label ownership conflicts, stale reports, missed uptake, and later recovery explicitly; avoid conflating all of them into one trace-level failure.
3. **Make applicability and observability explicit.** Preserve N/A for solo conditions and unknown for task-only records rather than counting their false booleans as clean negatives.
4. **Add independent labels and harder negatives.** Use two blind human reviewers with adjudication, including verification-heavy successful teams, genuinely ignored information, and abandoned handoffs. Evaluate per-class precision/recall only after those labels exist.
5. **Run a controlled native comparison.** Same scanner code, prompts, role filtering, context caps, model configuration, and repeated seeds/samples; keep full generations, costs, and failures. Compare fixed-evidence judging separately from an end-to-end annotation/candidate rerun.

## Frozen source and result registry

| Ref | Condition | Frozen evidence | Prompts / new output |
|---|---|---|---|
| E1 | Bundle / 4 agents / r2 | Transcript / Baseline JSON | Original prompt / Sol replay |
| E2 | Bundle / 4 agents / r1 | Transcript / Baseline JSON | Original prompt / Sol replay |
| E3 | Bundle / 3 agents / r2 | Transcript / Baseline JSON | Original prompt / Sol replay |
| E4 | Merger / 3 agents | Transcript / Baseline JSON | Original prompt / Sol replay |
| E5 | Arithmetic / solo | Transcript / Baseline JSON | Original prompt / Sol replay |
| E6 | Arithmetic / P-E-V | Transcript / Baseline JSON | Original prompt / Sol replay |
| E7 | Task-only room | Transcript / Baseline JSON | Original prompt / Sol replay |
| E8 | GPQA / solo | Transcript / Baseline JSON | Original prompt / Sol replay |
| E9 | GPQA / P-E-V | Transcript / Baseline JSON | Original prompt / Sol replay |

The pinned baseline includes mixed historical prompt versions, source text caps, 11 operator posts, and no execution logs. This report preserves those facts rather than silently repairing the baseline.

Machine-readable artifacts: paired comparison, non-blind evidence audit, input checksums, replay provenance. Reproduction instructions are in README.md. The builder performs no model calls.

Prepared October 18, 2026. Descriptive, small-cohort evidence audit; no causal model improvement or calibrated error-rate claim.